PVC/DEHP 40% — Simulation Results Summary

Two correct force fields give 300 K densities within ~2% of Zhou & Milner (2018), and PVC and DEHP share a single MSD-based Tg near 350 K. The remaining gap is most likely a small-box effect, not a force-field error.

System and setup

Item	Value
Reference	Zhou & Milner, Macromolecules 2018, 51, 3865–3873
Composition	16 PVC 20-mers + 34 DEHP = 4196 atoms, 39.9 wt% DEHP (paper: 96 + 200, 39.4 wt%)
Box (cubic)	3.88 nm at 600 K, 3.60 nm at 300 K (paper: ~6.5 nm)
Force field A	LigParGen: OPLS-AA LJ + CM1A-LBCC charges (standard σ)
Force field B	Vanilla OPLS-AA: table charges and LJ, as in the paper
Software	GROMACS 2026.3, Apple Silicon GPU (OpenCL)
Common settings	dt 1 fs, h-bond LINCS, PME + LJ cut-off 1.2 nm (potential-shift), DispCorr EnerPres, V-rescale τt 0.2 ps
Slow cool (Tg run)	`new_annealing`: 600 → 300 K over 50 ns (6 K/ns), Parrinello–Rahman τp 2 ps, force field A
Quench cycles	600 K NPT → 300 K NPT (8 ns) → 600 K (5 ns) → …, C-rescale τp 5 ps, force field B

Density

Both valid force fields sit 1–2% below the paper in the 16/34 box.

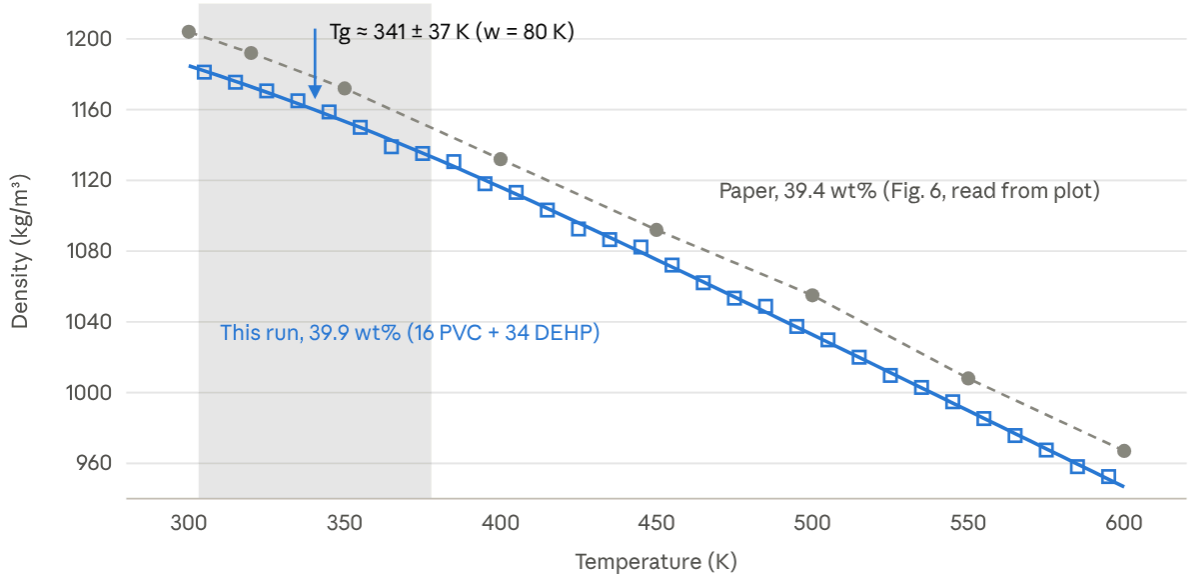
Run	Box	Protocol	600 K (kg/m³)	300 K (kg/m³)
Paper (vanilla OPLS-AA, Fig. 6, adjusted to 39.9 wt%)	96/200	slow cool 5 K/ns	~964	~1200
Vanilla OPLS-AA (`npt_full`, `tc1`)	16/34	600 K NPT → quench	946 ± 2–3 (1–5.4 ns)	1180 ± 2 (tc1, 2–8 ns)
LigParGen (`new_annealing`)	16/34	slow cool 6 K/ns	~950–953	~1183
LigParGen (`npt_300new_cycle3`)	16/34	quench cycles	—	~1207

Vanilla vs LigParGen differ by ~7 kg/m³ at 600 K (<1%), so the force-field choice matters less than box size. The paper's 300 K value was read from Fig. 6 marker centres (±~5 kg/m³).

Fitting the paper's Eq. 1 to the slow-cool density gives Tg ≈ 340 ± 37 K with the transition width fixed at 80 K.

Density-based Tg is ~340 K, but stopping at 300 K leaves it loosely bounded

Paper's Eq. 1 (w fixed at 80 K) fitted to 10 K bins of the LigParGen cool, 600 to 300 K at 6 K/ns



Slow-cool density binned in 10 K steps; paper values digitized from Fig. 6 of Zhou & Milner (2018)

The fit is loose because the cool stops at 300 K, leaving only ~40 K of glassy branch; the paper fitted 100–600 K. With w = 70 or 90 K, Tg moves to 355 or 322 K, and a free-w fit does not converge. The value agrees with the MSD-based Tg (~350 K) within error. Melt

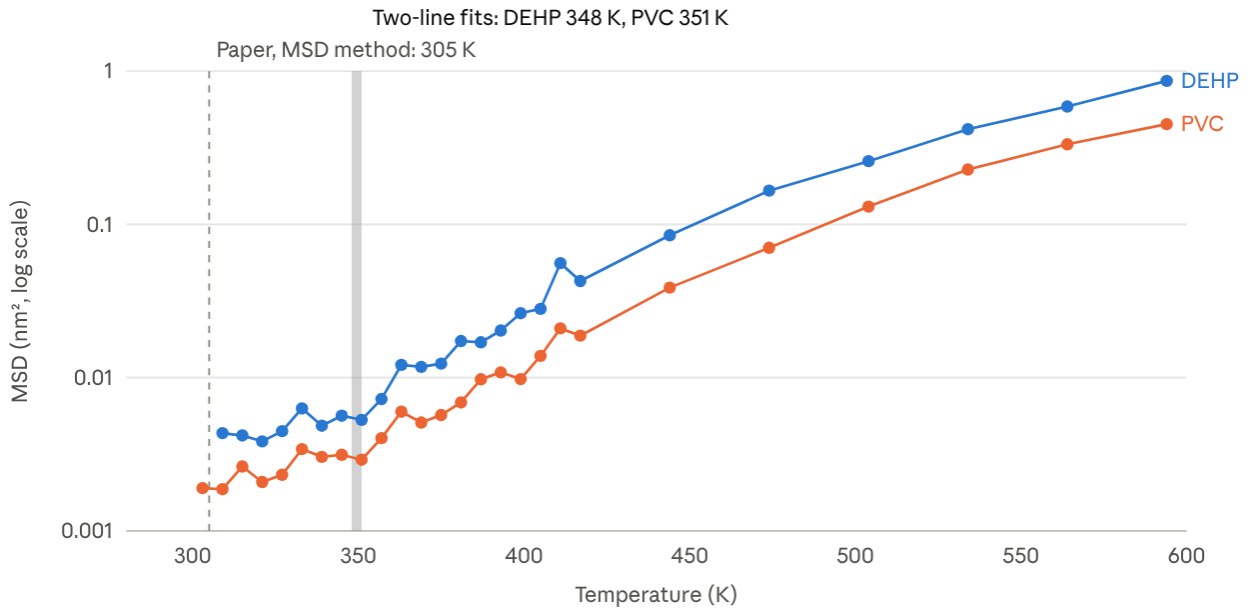
slope is −0.86 kg/m³/K.

Dynamics: MSD, Tg and diffusion

PVC and DEHP freeze together: two-line fits on log MSD(200 ps) give Tg ≈ 351 K for PVC (68% range 350–375 K) and ≈ 348 K for DEHP (337–352 K). DEHP stays 1.6–2.7× more mobile than the PVC chain centres at every temperature.

PVC and DEHP freeze together near 350 K; DEHP stays about 2x more mobile

Centre-of-mass MSD at 200 ps, LigParGen slow cool 600 to 300 K at 6 K/ns; 1 ns windows below 420 K, 2 ns above



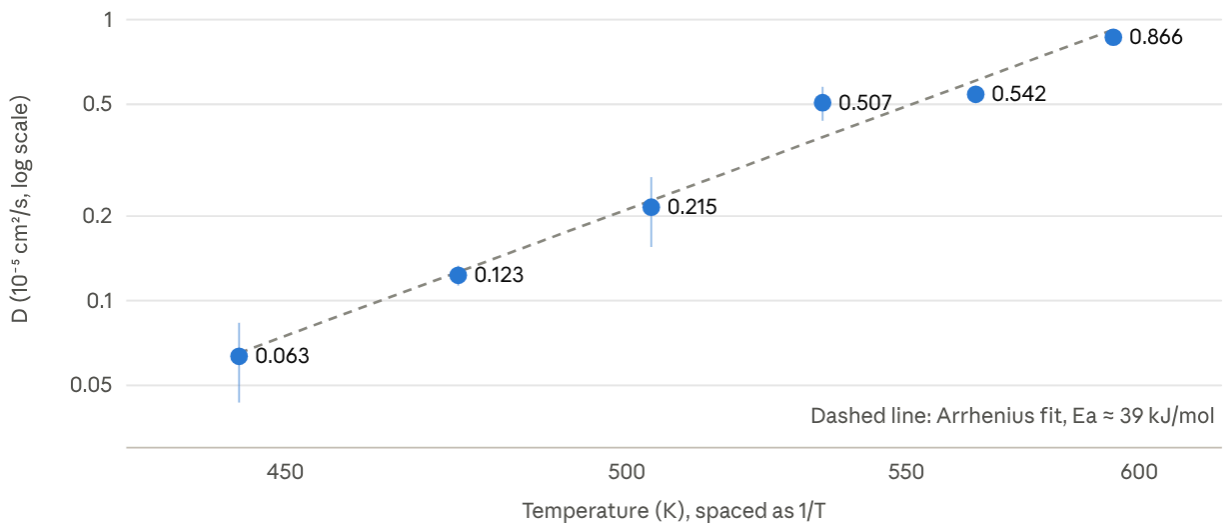
gmx msd on the slow-cool trajectory · 25 windows, 309–594 K

The fits use only the 1 ns windows below 420 K; adding melt points makes them chase the liquid branch. One shared Tg fits a well-mixed blend with no phase separation. The density-based fit in the Density section (~340 ± 37 K) agrees. Both values sit above the paper's 305 K (MSD) and ~316 K (density), possibly from the onset criterion, the LigParGen charges or the small box. Whole-molecule MSD cannot show the paper's local-Tg shift around DEHP; that needs layer-resolved analysis.

DEHP diffusion follows an Arrhenius law above 440 K, with an apparent activation energy of ~39 kJ/mol (R^2 = 0.98).

DEHP diffusion rises 14x from 444 to 594 K, with an Arrhenius slope

gmx msd fit 300–1200 ps on 2 ns windows of the slow cool; x axis spaced as 1/T



gmx msd -mol diffusion coefficients · 6 windows, 444–594 K

Below ~440 K the MSD does not reach the diffusive regime within 2 ns, so no D is reported there. A small periodic box slows diffusion (Yeh–Hummer, ∝ 1/L), so these values are likely underestimates. Error bars are gmx's two-half estimates; ±0.00 at 564 and 594 K means the halves agreed, not zero error.

Chain conformation (PVC 20-mer)

Ree matches the paper; Rg is ~10% smaller, plausibly because chains span the small box.

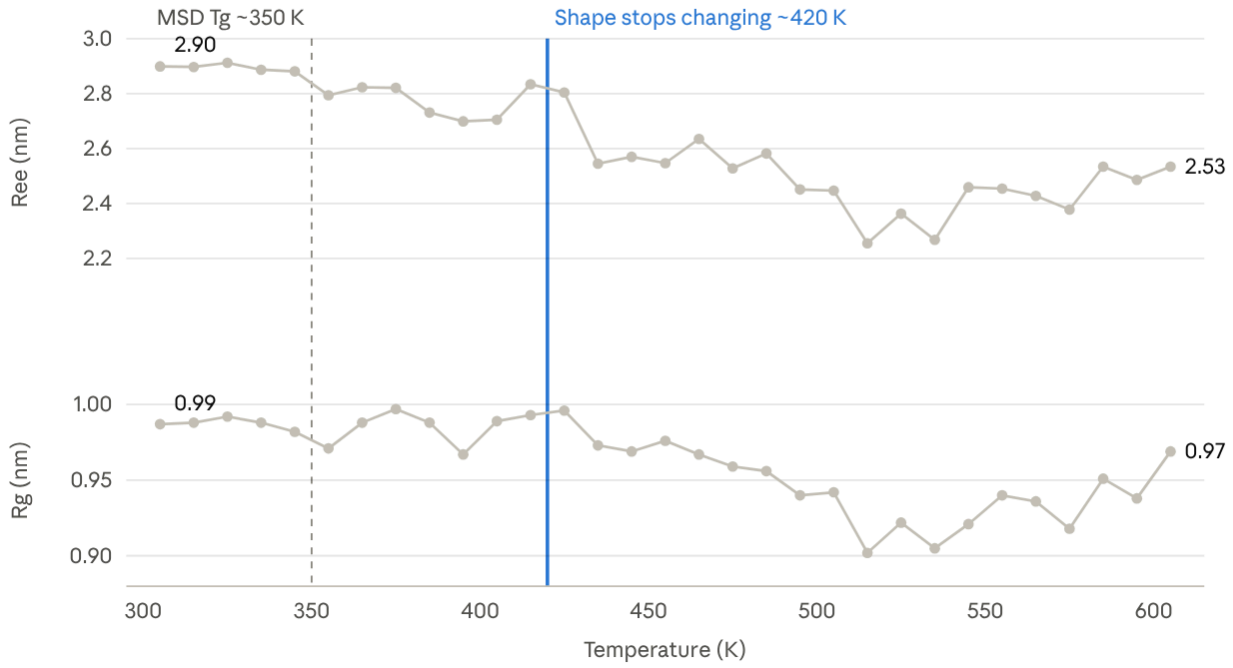
Property	Value	Method	Paper (Fig. 1)
Ree, last frame (300 K)	2.88 ± 0.65 nm (spread between 16 chains; ±0.16 on the mean)	`chain_ree.py`, backbone C00→C28	~2.9 nm
Rg, last frame (300 K)	0.99 ± 0.10 nm	`chain_ree.py`	~1.1 nm
Rg, averaged 600 → 300 K	1.015 nm	`gmx polystat`	—
Ree/Rg	~2.9 (Gaussian 2.45, rod 3.46)	—	—
Largest Ree seen	4.57 nm (full stretch ~4.9 nm)	`chain_ree.py`, 501 frames	—

The largest Ree (4.57 nm) exceeds the 300 K box (3.60 nm), so some chains span the box and can meet their own periodic image. `gmx polystat` reports Ree = 0.109 nm, which is a C–H bond: it uses the itp's first and last atoms, so it cannot give Ree for this atom ordering.

On cooling, Ree grows ~20% (≈ 2.4 → 2.9 nm) and Rg ~6% (≈ 0.93 → 0.99 nm) as the backbone shifts from gauche to trans (trans fraction 0.68 → 0.72).

PVC chains stretch on cooling and stop changing shape below ~420 K

Mean over 16 chains in 10 K bins, LigParGen slow cool 600 to 300 K at 6 K/ns (chain_ree.py)



Per-chain end-to-end distance and radius of gyration, binned in 10 K steps · 31 bins, 605–305 K

Chain shape freezes near 420 K, well above the ~350 K MSD Tg, because whole-chain rearrangement is far slower than centre-of-mass rattling. The 300 K Ree and Rg therefore reflect the cooling history, not equilibrium at 300 K. Bin-to-bin noise is about ±0.1 nm in Ree.

Fluctuations, structure and other values

The vanilla 300 K glass (tc1) gives $\kappa T \approx 4.5 \times 10^{-10}$ Pa^{-1} from 6 ns of C-rescale volume fluctuations; αP and Cp from the same run are less reliable (notes below the table).

Quantity	Value	Source
⟨V⟩, 300 K	46.870 ± 0.063 nm³ (box 3.606 nm)	vanilla tc1, 2–8 ns, 60,001 points
κT, 300 K	4.5×10^{-10} Pa^{-1} (≈ 4.0×10^{-10} after removing volume drift)	vanilla tc1, `gmx energy -fluct_props`
Bulk modulus, 300 K	2.2 GPa	1/κT, vanilla tc1
αP, 300 K	6.2×10^{-4} K^{-1} (uncertain)	vanilla tc1, V–H cross-fluctuations

Quantity	Value	Source
Cp, 300 K	2.85 J/(g K) (classical; experiment ~1.5–1.8)	vanilla tc1, enthalpy fluctuations
Volume drift, 300 K	−0.34 nm³ over 6 ns (−0.7%, slow densification)	vanilla tc1
Density SD, 600 K	13–20 kg/m³	vanilla `npt_full`, 1 ns blocks
Volume, 600 K	58.3 nm³ (box 3.88 nm)	`new_annealing`, `npt_full`
Pressure	mean ~+25 bar over 20 ps, ±~1000 bar instantaneous (normal for this box)	vanilla `npt_full`
Cl–ester O RDF, first shell	~0.48 nm, g ≈ 0.86–0.89 (depleted)	40% cycle 3, RESP vs LigParGen nearly identical
Cl–ester O RDF, second shell	~0.75–0.78 nm, g ≈ 1.1	same

αP comes from slowly converging V–H cross-fluctuations and sits close to the melt value (~7.8×10^{-4} K^{-1}), so it needs checking against other glasses. Classical MD gives every vibration full kBT, so Cp is overestimated.

RDFs should be cut at L/2 ≈ 1.8 nm; the 25% file drops to g ≈ 0.82 near 1.6 nm, which is a box-edge artifact.